LLM PREPROCESSING COMPARISON REPORT

Baseline vs Preprocessed Model Performance Analysis

Experiment ID	80edfb3361f8
Generated at	2026-11-09 15:42:52
Total records	7
Models tested	4
Queries tested	1
Successful responses	7
Failed responses	0

1. Model-wise Performance Summary

Model	Condition	Total	Successful	Failed	Avg latency (s)	Avg response chars
llama	preprocessed	1	1	0	21.489	617.0
qwen	baseline	1	1	0	14.328	741.0
qwen	preprocessed	1	1	0	8.112	719.0
smollm2	baseline	1	1	0	13.751	587.0
smollm2	preprocessed	1	1	0	14.048	508.0
tinyllama	baseline	1	1	0	13.446	707.0
tinyllama	preprocessed	1	1	0	8.577	668.0

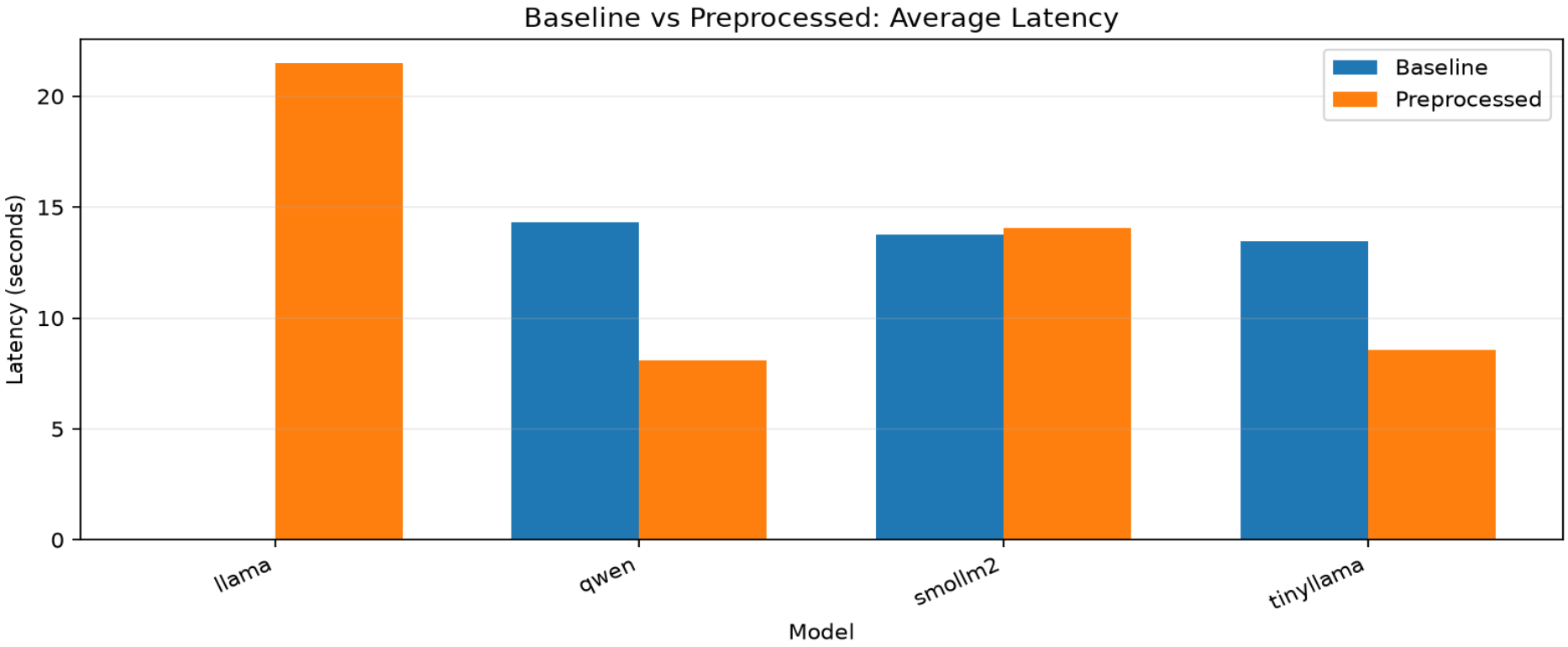
2. Baseline vs Preprocessed Comparison

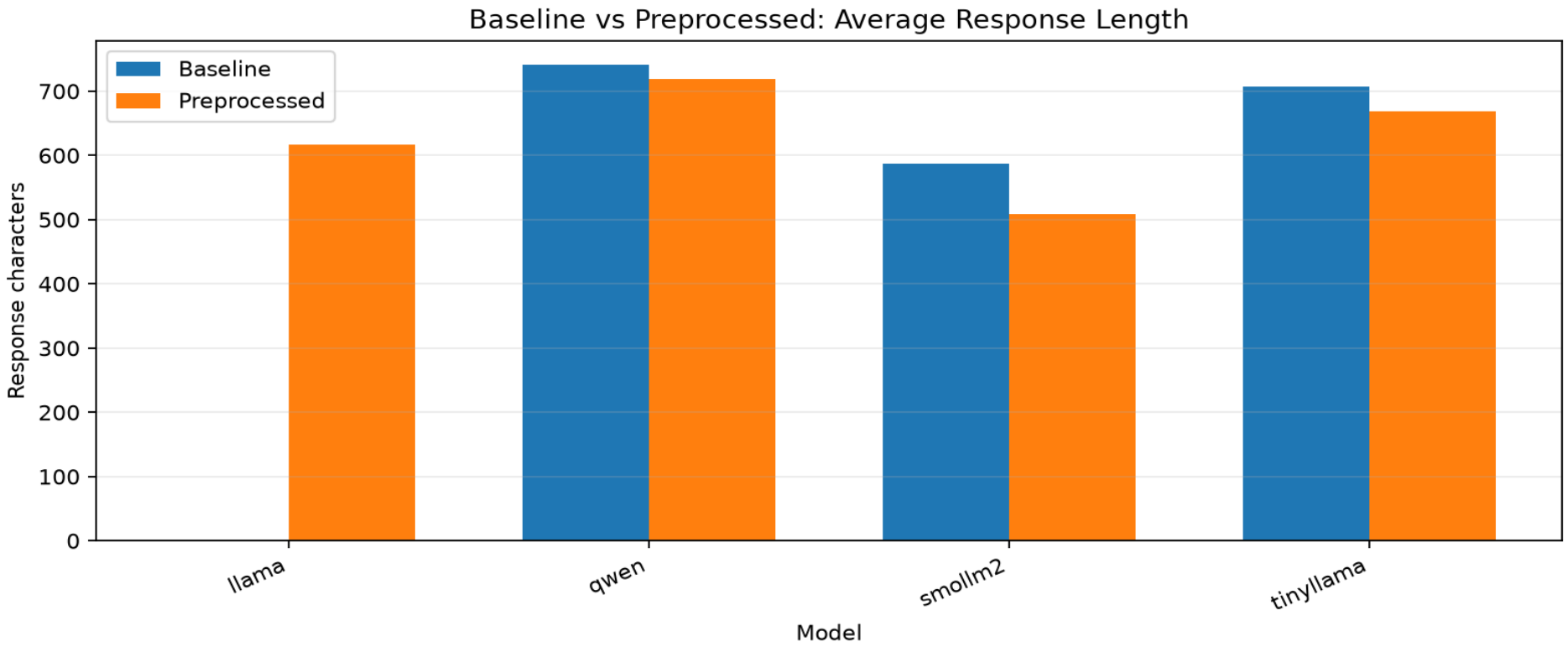
Model	Paired queries	Baseline latency (s)	Preprocessed latency (s)	Latency change (s)	Baseline chars	Preprocessed chars	Length change
qwen	1	14.328	8.112	-6.215	741.0	719.0	-22.0
smollm2	1	13.751	14.048	+0.297	587.0	508.0	-79.0

Model	Paired queries	Baseline latency (s)	Preprocessed latency (s)	Latency change (s)	Baseline chars	Preprocessed chars	Length change
tinyllama	1	13.446	8.577	-4.868	707.0	668.0	-39.0

Successfully paired model-query comparisons: 3. Negative latency change means preprocessing was faster. Positive latency change means preprocessing was slower. Negative response-length change means the response was shorter.

3. Performance Comparison Charts





4. Failed Requests and Error Details

No failed requests were recorded.

5. Interpretation and Evaluation Notes

- Latency values are averaged over successful responses only.
- Failed requests are reported separately.
- Baseline and preprocessed comparisons are paired using the same model and original query.
- Response length measures the number of characters returned; it does not establish answer correctness.
- A shorter response or lower latency does not automatically mean better response quality.
- Quality evaluation requires a documented rubric or validated benchmark. No factual accuracy score is inferred by this report.